



Superblocks and Electoral Outcomes: Examining Local and Spatial Spillovers in Barcelona

Dr. Jaime Orrego-Oñate,
MBF Nuñez, A. Guevara, O. Marquet

Motivation

- Cities around the world, aiming to reduce personal vehicle travel, are **repurposing space from cars to active transportation**.
- However, **fears of electoral backlash**, driven by strong and vocal opposition, can **suppress or delay these efforts**.
- But is the backlash real—or just a loud minority?



Aim

- Use **longitudinal electoral data**.
 - Analyze how **electoral behavior changes** before and after an intervention.
 - Since interventions are not implemented everywhere, apply a **difference-in-differences** approach using econometric methods.
-



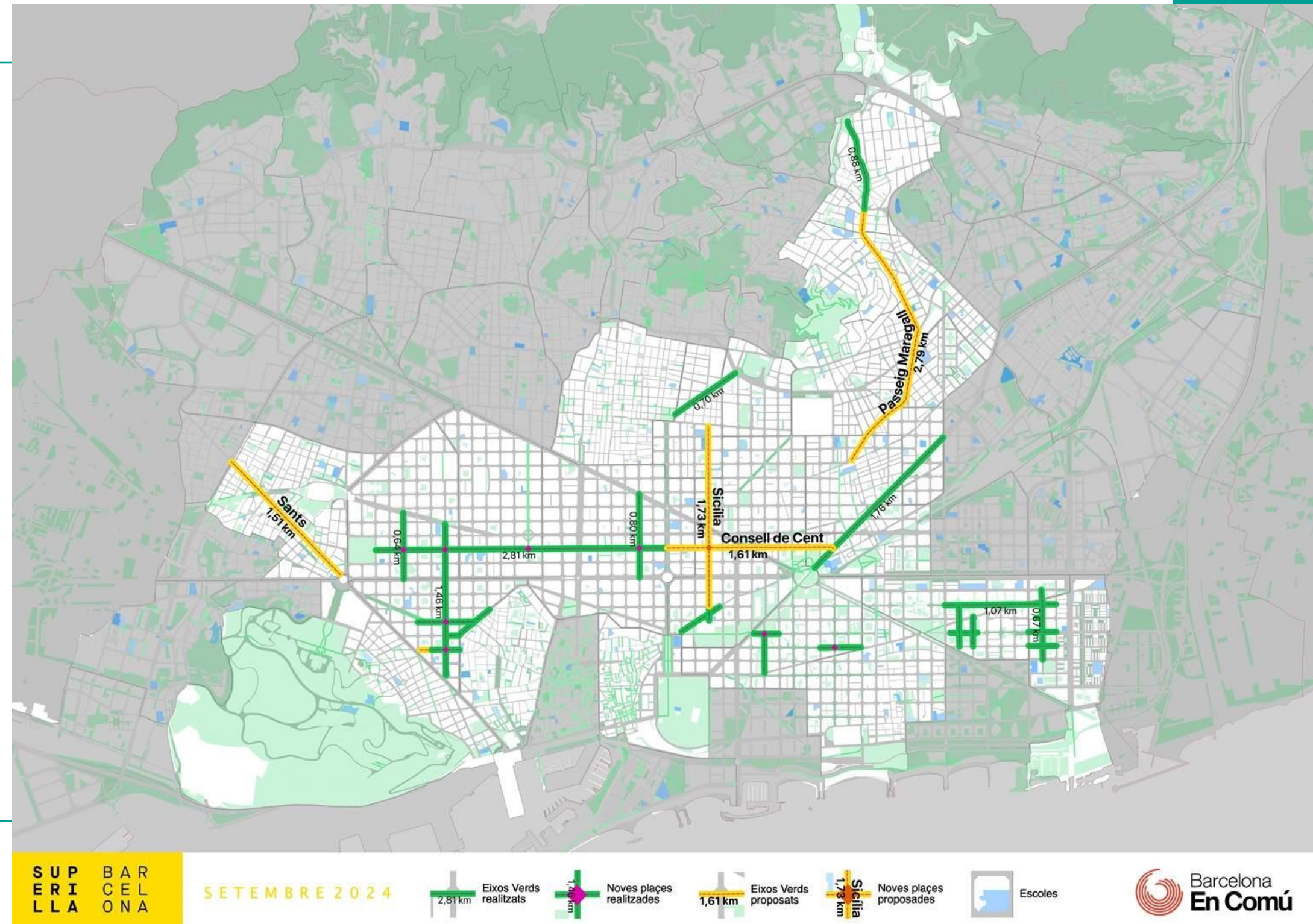
Challenges

- Electoral behavior does not follow a single causal path.
- People's voting decisions can be influenced by a variety of factors.
- Therefore, **isolating the effect** of an urban intervention can be **challenging**.



Case Study: Superblocks in Barcelona

- Local governments that implemented Superblocks during the periods 2015–2019 and 2019–2023.
- Superblocks are essentially the pedestrianization or semi-pedestrianization of streets.





Green axes as part of the superblocks





Method

- Electoral results at the census tract level for 2015, 2019, and 2023
 - The system is proportional, so there is no direct mayoral win; rather, a coalition is formed to elect the mayor
 - In 2015 and 2019, Barcelona en Comú led the coalition and held the mayoralty. In 2023, the Partit dels Socialistes de Catalunya led the coalition (in partnership with BeC)
 - **Test different econometric approaches** to estimate the **effect of Superblock** implementation on Barcelona en Comú's **vote share at the zonal level**
-



Econometrical approach

- Test the effect of Superblock interventions using longitudinal models at the zonal level
- Model specification Voting Share of BeC (ruling party - %)
 - Zonal income
 - Population density
 - National scale election results (same year at different date)
- Progressively increase model complexity:
 - **Simple linear regression**: baseline association between predictors and BeC vote share
 - **Fixed effects models**: control for time-invariant characteristics of each zone
 - **Spatial regression models**: account for spatial autocorrelation and spillover effects
- Assess the stability of the Superblock coefficient across model specifications



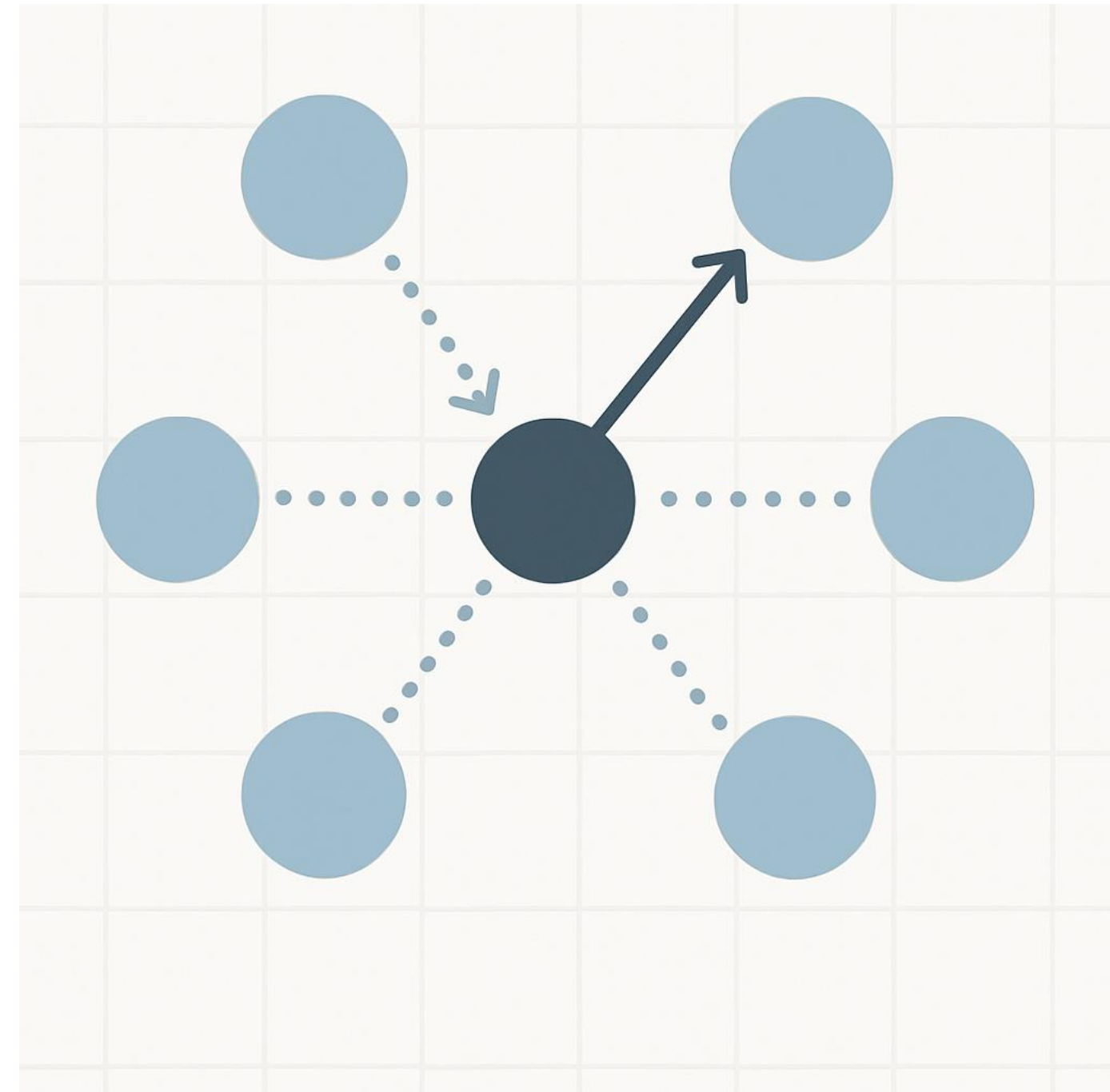
Results superblock effect over vote share

- In areas with **Superblock implementation**, the BeC voting share increased by approximately **1% to 2.5%**, depending on the model
 - All coefficients are **statistically significant**
 - After accounting for fixed effects, the Superblock **effect remains stable**, with an **estimated increase of around 1.7%** in BeC vote share
-

Linear Model	0.025
Fixed effects	0.017
Fixed effects (first difference)	0.012
Spatial Regression (lag version)	0.016
Spatial Regression (error version)	0.017

Spillovers

- Assuming spatial regressions are reliable, we can examine spillover effects.
- Direct effect: **1.67%**
- Indirect effect: **-0.3%**





Discussion

- Challenge:** Lack of a clear causal mechanism to identify valid correlations.
- Proposal:** Use an econometric approach to estimate stable effects.
- Findings:** Overall, there is a consistent vote share premium of ~1–2%, while any backlash appears to be smaller (~0.3%).

Next steps:

- Conduct **sensitivity analysis** with **additional covariates**.
 - Since **Superblocks are not randomly assigned**, explore the use of an instrumental variable to **estimate the likelihood of a Superblock** being implemented in a given census tract
-



Thanks!
jaime.orrego@uab.cat



UAB
Universitat Autònoma de Barcelona



TÍTULO

SUBTÍTULO



1 ÍNDICE PTMA

Desarrollar un indicador sintético del potencial territorial para la movilidad activa.

2 VISOR ON-LINE

Cartografiar el Atlas de la Movilidad Activa y hacerlo accesible al conjunto de la población.

3 FORMACIÓN

Formar al grupo de investigación en innovación y emprendimiento.

Ofrecer un programa de formación a entidades públicas y privadas.

TÍTULO

SUBTÍTULO

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed diam nonummy nibh euismod tincidunt ut laoreet dolore magna aliquam erat volutpat. Ut wisi enim ad minim veniam, quis nostrud exercitation ullamcorper suscipit lobortis nisl ut aliquip ex ea commodo consequat.

Duis autem vel eum iriure dolor in hendrerit in vulputate velit esse molestie consequat, vel illum dolore eu feugiat nulla facilisis at vero eros et accumsan et iusto odio dignissim qui blandit praesent luptatum zzril delenit augue dui dolore te feugait nulla

facilisi. Blandit praesent luptatum zzril delenit augue dui dolore te feugait nulla facilisi. Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed diam nonummy nibh euismod tincidunt ut laoreet dolore magna aliquam erat volutpat. Ut wisi enim ad minim veniam, quis nostrud exercitation ullamcorper suscipit lobortis nisl ut aliquip ex ea commodo consequat.

Duis autem vel eum iriure dolor in hendrerit in vulputate velit esse molestie consequat, vel illum dolore eu feugiat nulla facilisis at vero eros et accumsan et iusto

odio dignissim qui blandit praesent luptatum zzril delenit augue dui dolore te feugait nulla facilisi.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed diam nonummy nibh euismod tincidunt ut laoreet dolore magna aliquam erat volutpat. Ut wisi enim ad minim veniam, quis nostrud exercitation ullamcorper suscipit lobortis nisl ut aliquip ex ea commodo consequat. Duis autem vel eum iriure dolor in hendrerit in vulputate velit esse molestie consequat.
